Candidate's Seat No………….

CHAROTAR UNIVERSITY OF SCIENCE &TECHNOLOGY

Second Semester of M.Tech Examination (IT)

May 2018

IT746 Computational Intelligence

Date: 05.05.2018,Saturday **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks:70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 **Answer the following questions.** **[07]**

1. Match the following. [03]

A	Crossover	1	Bitwise changing
B	Mutation	2	Survivability
C	Selection of chromosomes	3	Sequences of designing a fuzzy logic machine
D	Fuzzification-Rule, Evaluation-Defuzzification	4	Many-valued logic in which the truth values
E	Fitness function	5	Individual genomes are chosen from a population
F	Fuzzy Logic	6	Recombination event occurs in between the genes.

2. Let $\tilde{A}_1$and $\tilde{A}_2$ are two fuzzy sets corresponding to variable temperature. [02]
Fuzzy set $\tilde{A}_1$ = {(x1,0.2),(x2,0.8),(x3,0.6)} , $\widetilde{A}_2$ = {(x1,0.2),(x2,0.6),(x3,0.9)}
Perform Disjunctive sum operation.

3. What are the objectives of soft computing? How it is differ from hard computing? [02]

Q-2 (a) Explain Fuzzy Inference System in detail with its block diagram. **[04]**

Q-2 (b) **Answer the following questions (Any Two)** **[10]**

1. Let $\widetilde{high}$ and $\widetilde{very\ high}$ are two fuzzy sets corresponding to variable temperature. Consider Universal set U={15,20,25,30,35,40,45,50}
Fuzzy set $\widetilde{high}$ = {0.4/15 + 0.6/20 + 0.8/25 + 1/30 + 0.5/35 + 0.2/40} and
$\widetilde{very\ high}$ = {0.4/20 + 0.6/25 + 0.8/30 + 0.8/35 + 0.9/40 + 1/45 + 1/50}
Find (i) union, (ii) intersection, (iii) complement and (iv) difference and (v) verify De Morgan's Law.

2. Consider the following population of binary strings for maximization problem, Evaluate fitness function according to f(x) = x^2, Index starts from 0.

String No.	Initial Population
x1	01101
x2	11000
x3	01000
x4	10011

Find out the expected number of copies of the best string in the above population of the mating pool under Roulette wheel selection, 1-point cross-over (String no. 1 and 2 at 3rd index and 3 & 4 at 1st index) after first iteration.

3. Consider a set P= {P1, P2, P3} of four varieties of paddy plants, set D = {D1, D2} of the various disease affecting the plants and S= {S1, S2, S3} be the common symptoms of the diseases. Let $\widetilde{R1}$ be a relation on P × D and $\widetilde{R2}$ be a relation D × S

$$\widetilde{R1} = \begin{bmatrix} 0.6 & 0.1 \\ 0.2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \widetilde{R2} = \begin{bmatrix} 0.7 & 0.4 & 0.7 \\ 0.5 & 0.8 & 0.9 \end{bmatrix}$$

Obtain the association of the plants with the different symptoms of the diseases using max-min composition.

4. Let X={a,b,c,d} & Y={p,q,r,s} and Fuzzy Sets

$\tilde{A}$={(a,0),(b,0.8),(c,0.6),(d,1)}

$\tilde{B}$={(p,0.2),(q,1),(r,0.8),(s,0)}

$\tilde{C}$={(p,0),(q,0.4),(r,1),(s,0.8)}

Determine the implication relations

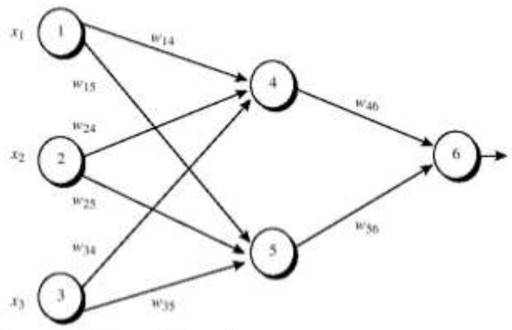
(i) IF x is $\tilde{A}$ THEN y is $\tilde{B}$.

(ii) IF x is $\tilde{A}$ THEN y is $\tilde{B}$. ELSE y is $\tilde{C}$

Q-3 Answer the following questions (Any Two) [14]

1. What is defuzzification? List out three different defuzzification methods and also explain with suitable example.
2. List out reproduction techniques in GA and explain Roulette-wheel selection with suitable example.
3. Perform a complete forward & backward step of the feed forward network 3-2-1 architecture using back propagation algorithm. Take below fig.4 as input and train the given network for output node3 and calculate new values of weight w46 & bias (theta6). Assume, target output=0.9, η=0.25,α=0.0001 and sigmoid function as below.

$$\varphi(v) = \frac{1}{1+\exp(-v)}$$



Initial input, weight, and bias values.

x_1	x_2	x_3	w_{14}	w_{15}	w_{24}	w_{25}	w_{34}	w_{35}	w_{46}	w_{56}	θ_4	θ_5	θ_6
1	0	1	0.2	−0.3	0.4	0.1	−0.5	0.2	−0.3	−0.2	−0.4	0.2	0.1

SECTION – II

Q-4 Answer the following questions. [07]

1. Define confusion matrix. List out measurement rates of predictive model and explain any three measurement rates with suitable example. [03]
2. Explain reinforcement learning technique with neat sketch. [02]

3. Give the comparison between supervised learning and unsupervised learning with neat sketch. [02]

Q-5 (a) What is Particle swarm optimization? Give the difference between PSO and GA. **[04]**

Q-5 (b) **Answer the following questions (Any Two)** **[10]**

1. What is crossover in GA? Explain crossover rate and mutation rate.
2. List out encoding techniques in Genetic Algorithm and explain any four encoding techniques with suitable example.
3. Determine optimal hyper plane for SVM model y = wx + b considering given augmented vectors s1= [1 1 1] s2= [1 -1 1] s3= [3 -1 1] s4=[3 1 0] using linear support vector machine algorithm.
4. Demonstrate and model given real dataset "Auto Insurance in Sweden" using linear regression algorithm.

X	Y
1	3
2	3
4	3
3	3
5	3

Q-6 **Answer the following questions (Any Two)** **[14]**

1. Consider the following blue labeled data and red labeled data respectively.

Blue class vectors are: $\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}-1\\1\end{pmatrix}, \begin{pmatrix}-1\\-1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}$

Red class vectors are: $\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}, \begin{pmatrix}-2\\0\end{pmatrix}, \begin{pmatrix}0\\-2\end{pmatrix}$

And use below mapping function for non linear support vector machine. Determine optimal decision boundry and find out w and b parameter for SVM y=w.x+b

$$\Phi\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{cases}\begin{pmatrix}6 - x_1 + (x_1 - x_2)^2\\6 - x_2 + (x_1 - x_2)^2\end{pmatrix} & \text{if } \sqrt{x_1^2 + x_2^2} \geq 2\\ \begin{pmatrix}x_1\\x_2\end{pmatrix} & \text{otherwise}\end{cases}$$

2. Consider following eight points.P1 (2, 2), P2 (1, 14), P3 (10, 7), P4 (1, 11), P5 (3, 4), P6 (11, 8), P7 (4, 3), P8 (12, 9) Take P1, P2, and P7 as initial centroids. Then apply k-means clustering algorithm and Euclidean distance to find the three cluster centers after the second iteration.

3. Discover rule(s) using basic Ant-Miner algorithm of Ant Colony Optimization for below table.

OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	PLAY
sunny	85	85	false	Don't Play
sunny	80	90	true	Don't Play
overcast	83	78	false	Play
rain	70	96	false	Play
rain	68	80	false	Play
rain	65	70	true	Don't Play
overcast	64	65	true	Play
sunny	72	95	false	Don't Play
sunny	69	70	false	Play
rain	75	80	false	Play
sunny	75	70	true	Play
overcast	72	90	true	Play
overcast	81	75	false	Play
rain	71	80	true	Don't Play

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